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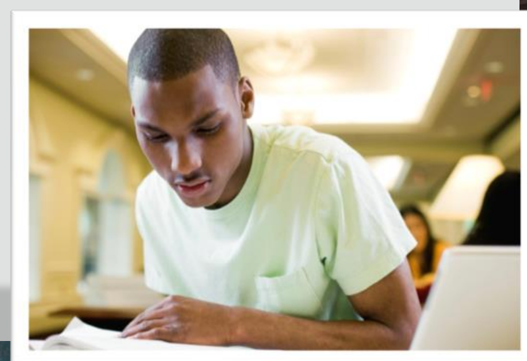
Database Foundations

5-2
Mapping Primary and Foreign Keys



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Objectives

- This lesson covers the following objectives:
- Decide on naming conventions for:
  - Primary key constraint names
  - Foreign key constraint names
  - Foreign key column names

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Objectives

- This lesson covers the following objectives:
  - Use Oracle SQL Developer Data Modeler to apply naming standards that map:
    - UIDs to primary key constraints
    - Relationships to foreign key columns and constraints
  - Map subtypes to tables

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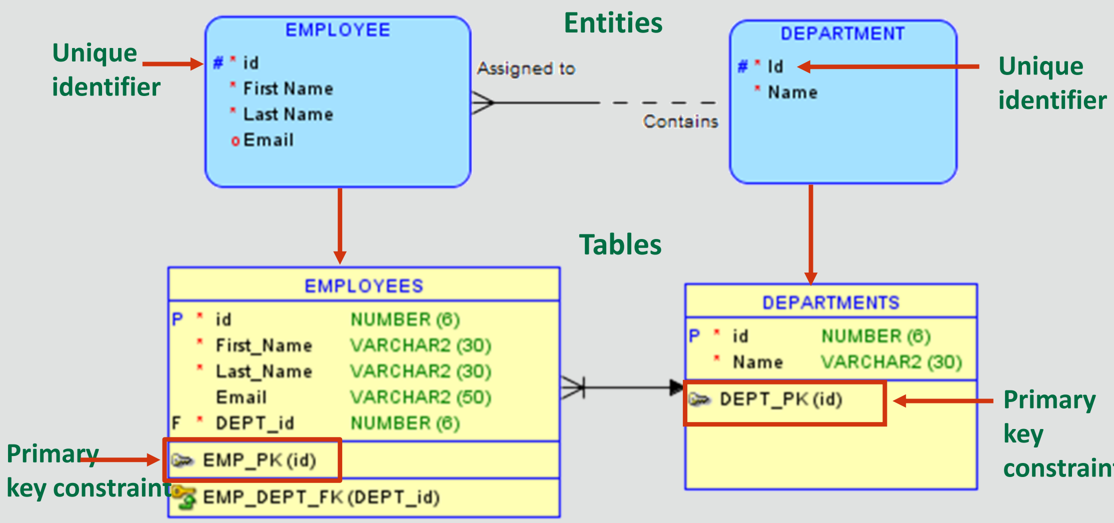
Naming Conventions

- Constraints enforce rules and ensure the consistency and integrity of the database. (Constraints are covered in more detail later in the course)
- Constraints should be given meaningful names to allow them to be easily referenced

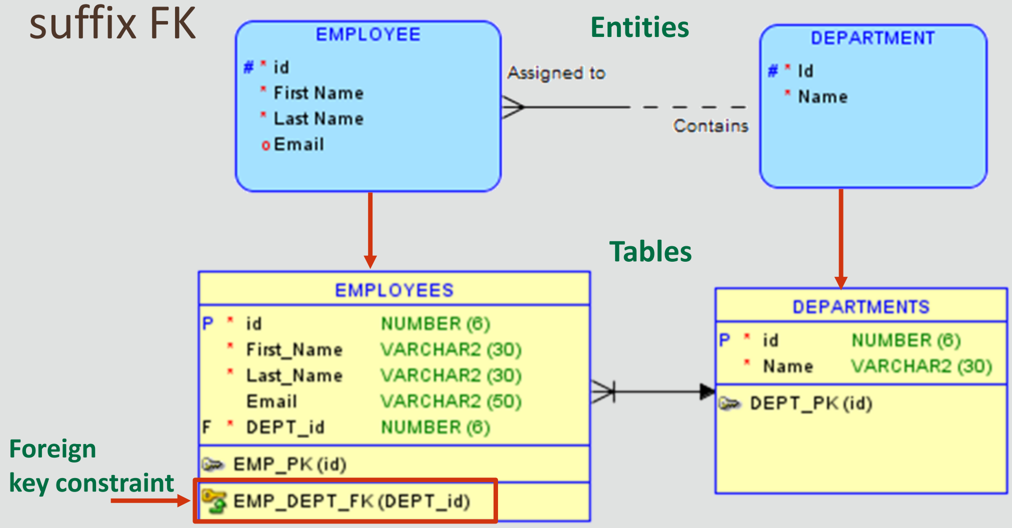
Naming Conventions for Primary Key Constraints

- Primary key constraints are named using the short table name, an underscore and the suffix PK



Naming Conventions for Foreign Key Constraints

- Foreign key constraints are named using the short table names of both tables, an underscore and the suffix FK



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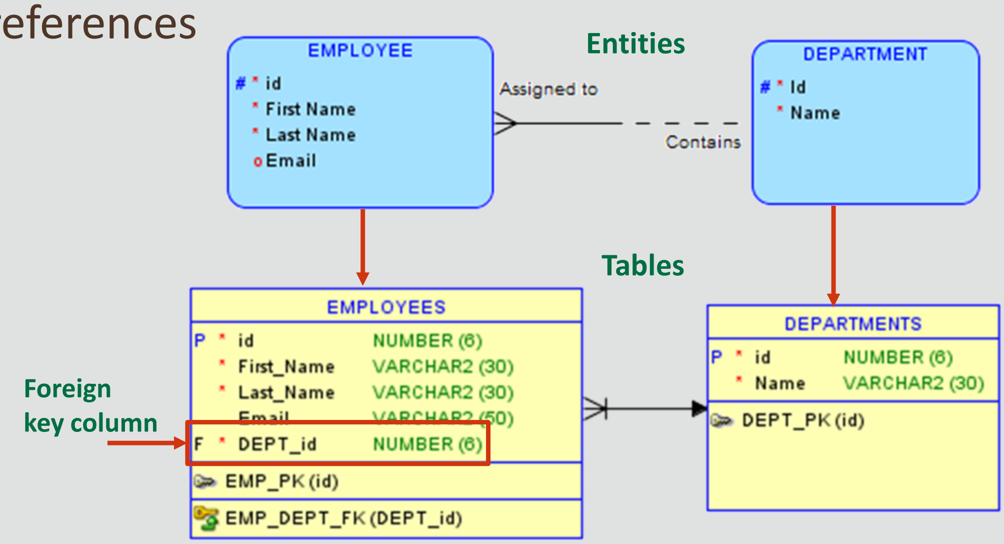
Naming Conventions for Foreign Key Columns

- Foreign key columns are named using the short table name and column name of the table the foreign key references



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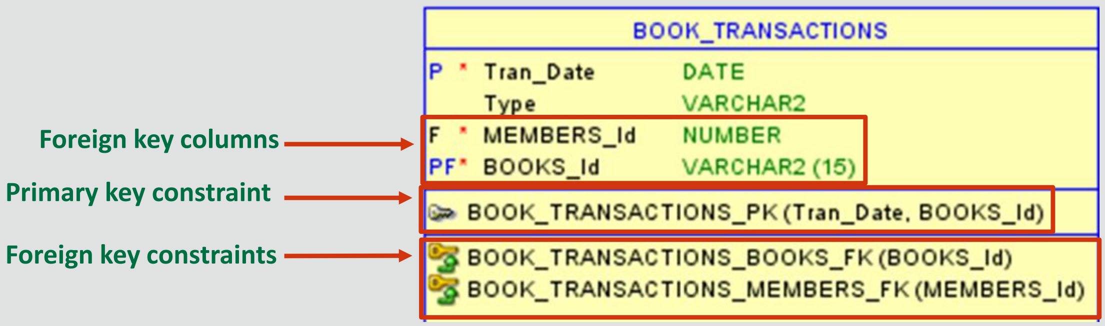
Apply naming standards in Oracle SQL Developer Data Modeler

- By default, constraint names are created using the full table name in Oracle SQL Developer Data Modeler. This can lead to constraint names that are very long, difficult to manage, and may exceed the maximum permitted number of characters in SQL



Create Name Abbreviations

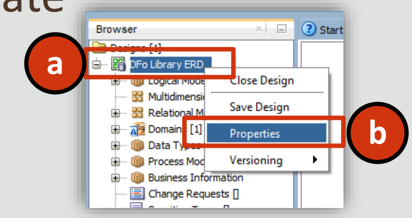
- To apply naming standards to constraint names, firstly create a .csv file in a spreadsheet application
- In the first column specify the full (plural) table names, and in the second column the abbreviation to use. Save as .csv file and note the location

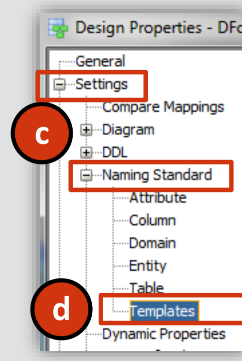
	A	B
1	PUBLISHERS	PUB
2	BOOKS	BK
3	AUTHORS	ATHR
4	MEMBERS	MEM
5	TRANSACTIONS	TRN

Example of a .csv file content

Defining Naming Templates

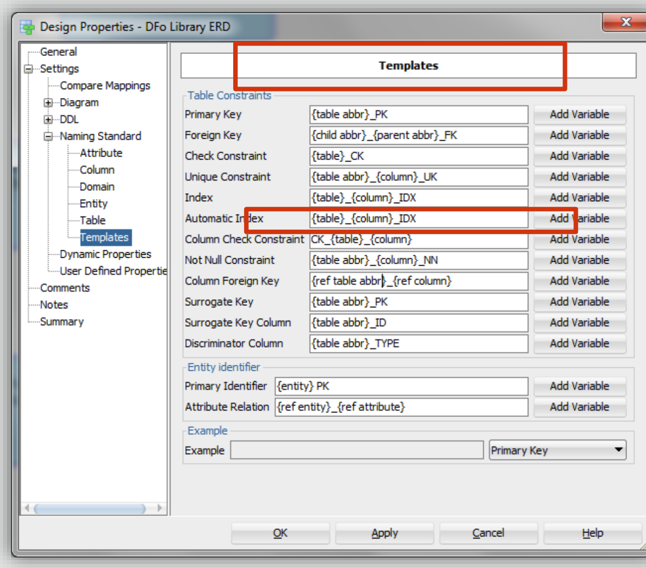
- To apply our abbreviations to our model, we need to set the pattern in the Naming template
- To access the Templates page:
  a. Right-click the name of the Design in the Object Browser
  b. Select Properties
  c. expand Settings and Naming Standard
  d. Select Templates





Defining Naming Templates (cont.)

- Edit the template to use abbreviated names for Primary and Foreign keys and Column Foreign Key by adding abbr to the pattern

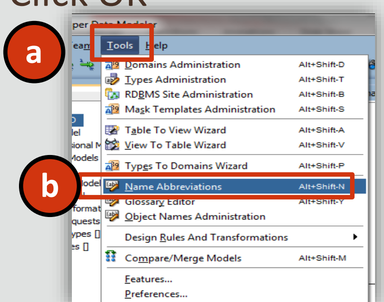


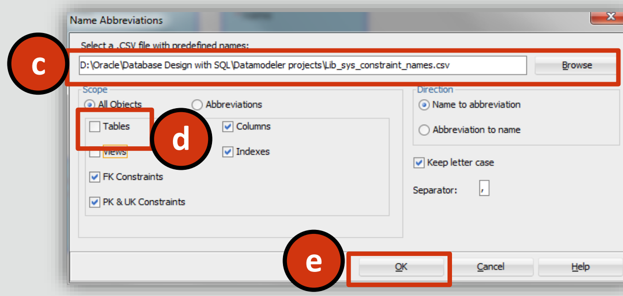
You can define templates (name patterns) for keys, indexes, and constraints by using combinations of predefined variables. The predefined variables include the following:

- {table}
- {table abbr}
- {child}
- {child abbr}
- {parent}
- {parent abbr}
- {column}
- {column abbr}
- {ref column}
- {ref column abbr}
- {seq nr}
- {model}
- Alphanumeric constants

Applying Naming Abbreviations

- To apply the abbreviations from the template
  - a. Select Tools
  - b. Select Name Abbreviations
  - c. Browse to the .csv file containing the abbreviations
  - d. Un-check Tables to maintain existing names from the Glossary
  - e. Click OK





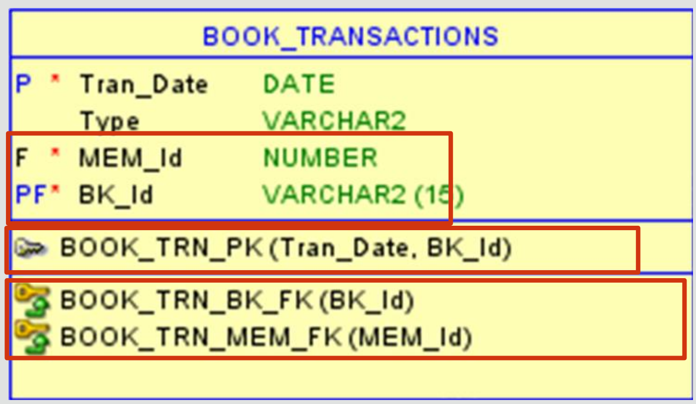
Applying Naming Abbreviations (cont.)

- The Relational model is now engineered with constraint names that follow our naming standards
  - Note: if the Logical model has already been engineered, sometimes it may be necessary to delete all objects in the relational model, re-engineer the logical, then apply the naming abbreviations

Foreign key columns

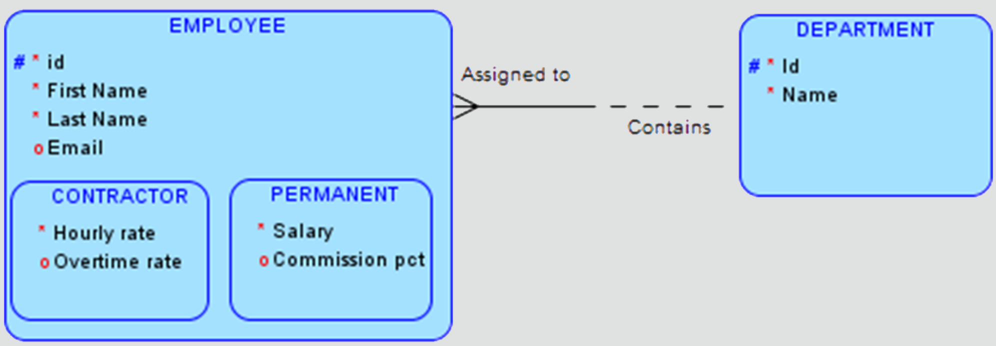
Primary key constraint

Foreign key constraints



Mapping Subtypes to Tables

- As you learned in a previous lesson subtypes can be mapped to tables in a number of ways
  - One table (single table implementation)
  - Two table (table for each child (subtype) implementation)



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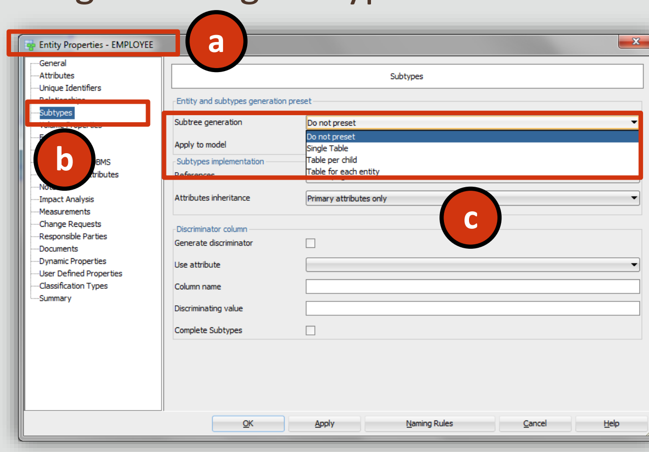
Mapping Subtypes to Tables (cont.)

- To select how subtypes are engineered in Oracle SQL Developer Data Modeler:
  a. Double click the Super type entity to edit properties
  b. Select Subtypes
  c. Select required method for generating subtypes from the Subtree generation dropdown box options



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Summary

- In this lesson, you should have learned how to:
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